

Use Case	Comment
Use Case Name	Create Task
Scope	Personal task management system
Level	“user-goal”
Primary Actor	Individual user
Stakeholders and Interests	User wants to create a new task to stay organized, meet deadlines, have structure, and organize tasks within projects and tags.
Preconditions	User must exist (be registered), authenticated, and logged into the system.
Success Guarantee	The task is created and saved into the system, optional associations of project and tags are formed, the initial status is recorded as “open”, and activity history is recorded.
Main Success Scenario	1. The user creates a new task and provides the task details 2. If a recurrence pattern is provided, the user specifies the interval (daily, weekly, or monthly) and the custom interval start and end dates. 3. The system verifies the inputs (including title uniqueness and date validity), creates the task template, and generates the sequence of occurrences. 4. The system creates the task, assigns the task with a unique Id, sets the status to “open”, and records the creation date. 5. System returns the task details. 6. User optionally assigns the task to a project 7. System verifies the project exists and forms the association. 8. User optionally add one or more tags to the task. 9. The system verifies tag name and associates the tag with the task. 10. The user optionally creates subtasks. 11. The system creates and links the subtask to the parent task.
Extensions	1a. Missing Title: 1. System detects the title is empty. 2. System informs the User that the title is required and halts creation. 1b. Invalid Priority: 1. System detects the priority level is not recognized/invalid. 2. System notifies the User to select a valid priority. 1c. Invalid Due Date: 1. System detects the due date is invalid (before the current date) 2. System notifies the User to select a valid date. 2a. Missing Recurrence Pattern: 1. The system detects the recurrence pattern intervals (daily, weekly, monthly) is empty. 2. System informs the User that a recurrence pattern interval is required and halts creation. 2b. Missing start and/or end date 1. The system detects the recurrence pattern start and/or end date are empty. 2. System informs the User that a start and end date are required and halts creation. 3a. Invalid Recurrence Pattern interval: 1. System detects the recurrence pattern interval is not a recognized value.

	- System notifies the user to select a valid recurrence pattern interval. 3b. Invalid start and/or end date - System detects start and end date for the recurrence pattern is invalid (start date is before current date or end date is before start date) - System notifies the user to select a valid start and/or end date. 3c. Non-unique Task Occurrence: - System detects that an occurrence with the same task name and due date already exists. - System notifies the User of the conflict and asks them to modify either the title or the recurrence start date. 3d. Occurrence generation error: - System determines that the chosen interval and dates would result in zero occurrences (ex: "Weekly on Mondays" but the start/end date range is only a Tuesday through Thursday). - System informs the User that no occurrences can be generated with those settings. 4a. Project Not Found: - System cannot find a project matching the provided project ID. - System informs the User that the project was not found, and the task remains unassociated with a project. 6a. Invalid Tag(s): - System detects tags that do not meet naming conventions or not found in the system. - System notifies the user about invalid tag and rejects the specific tag. 8a. Missing Subtask Title: - System detects the subtask title is empty. - System informs the User that the subtask requires a title and halts creation.
Special Requirements	Usability: The system should be able to give visual feedback to the user when creating a task within 20 seconds. Data Integrity: In case the system crashes when recording an activity history entry, the whole creation of the task should be undone so that the history can be reflective of the task state always. Accessibility: The task creation interface should be entirely navigable using keyboard shortcuts in order to allow power users to input data fast. Storage/Permanence: The system should be in a position to store infinite number of tasks and the related activities (projects and subtasks) without minimizing search performance since all tasks are present in the system regardless of status.